Chapter 8S. Learning Error and Neural Network Approximation

Supplementary handout: risk decomposition, curse of dimensionality, surrogate losses, universal approximation and Barron rates

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Remark 8S.1: about this chapter

A supplementary handout for two post-midterm lectures: sources of error (Sections 1–3), approximation (4–5), rate in the width (6–7). Sections 4–7 overlap Chapter 8 and are kept short.

1 Risk and the Sources of Learning Error

Motivation. Finite sample, imperfect optimizer and restricted class each cost excess risk.

1.1 Population and empirical risk

Definition 8S.2: population and empirical risk

Let $S = ((X_i, Y_i))_{i=1}^n \sim P^{\otimes n}$ be a training sample and ℓ a loss. For a predictor f,

$$R(f) = \mathbb{E}_{(X,Y)\sim P}\, \ell(f(X), Y), \qquad \widehat{R}_S(f) = \frac{1}{n}\sum_{i=1}^{n} \ell(f(X_i), Y_i). \tag{1}$$

R (**population risk**) averages over a fresh observation; $\widehat{R}_S$ is the **empirical risk**.

For a constrained class $\mathcal{F}_\delta \subseteq \mathcal{F}$, output $\widehat{f}_S \in \mathcal{F}_\delta$, and finite risks,

$$R(\widehat{f}_S) - \inf_{f\in\mathcal{F}} R(f) = \underbrace{\inf_{f\in\mathcal{F}_\delta} R(f) - \inf_{f\in\mathcal{F}} R(f)}_{\varepsilon_{\text{app}}} + \underbrace{R(\widehat{f}_S) - \inf_{f\in\mathcal{F}_\delta} R(f)}_{\varepsilon_{\text{est}}}. \tag{2}$$

Approximation error (cost of the restriction) and **estimation error** (distance to the best in class) are both ≥ 0 since $\mathcal{F}_\delta \subseteq \mathcal{F}$ and $\widehat{f}_S \in \mathcal{F}_\delta$.

Definition 8S.3: generalization gap and uniform deviation

$$\text{gen}_S(f) = R(f) - \widehat{R}_S(f), \qquad \Delta_S(\mathcal{F}_\delta) = \sup_{f\in\mathcal{F}_\delta} |R(f) - \widehat{R}_S(f)|. \tag{3}$$

The gap is signed, for one predictor; $\Delta_S \geq 0$ controls the whole class, so $|\text{gen}_S(\widehat{f}_S)| \leq \Delta_S(\mathcal{F}_\delta)$.

Remark 8S.4: notation

The lecture's ε_{gen} (a supremum) is Δ_S here, distinct from gen_S. n: sample size, d: input dimension, N: width.

1.2 Optimization error and a uniform bound

Definition 8S.5: optimization error

$$\varepsilon_{\text{opt}} = \widehat{R}_S(\widehat{f}_S) - \inf_{f\in\mathcal{F}_\delta} \widehat{R}_S(f) \;\geq 0. \tag{4}$$

It is zero exactly for an empirical risk minimizer. Adding and subtracting $\widehat{R}_S(\widehat{f}_S)$ and $\inf_{\mathcal{F}_\delta} \widehat{R}_S$,

$$\varepsilon_{\text{est}} = \text{gen}_S(\widehat{f}_S) + \varepsilon_{\text{opt}} + \Big(\inf_{f\in\mathcal{F}_\delta} \widehat{R}_S(f) - \inf_{f\in\mathcal{F}_\delta} R(f)\Big). \tag{5}$$

The first and third terms can be negative; the third cannot be dropped in general.

Theorem 8S.6: excess-risk bound

For $\widehat{f}_S \in \mathcal{F}_\delta$,

$$R(\widehat{f}_S) - \inf_{f\in\mathcal{F}} R(f) \le \varepsilon_{\text{app}} + 2\Delta_S(\mathcal{F}_\delta) + \varepsilon_{\text{opt}}.$$

Proof. Write $\Delta = \Delta_S(\mathcal{F}_\delta)$.

Step 1 (two-sided comparison). For every $f \in \mathcal{F}_\delta$,

$$R(f) - \Delta \le \widehat{R}_S(f) \le R(f) + \Delta.$$

Step 2 (the infimum term). Taking infima, $\inf R - \Delta \le \inf \widehat{R}_S \le \inf R + \Delta$: the last term of (5) is in $[-\Delta, \Delta]$.

Step 3 (the gap term). $\text{gen}_S(\widehat{f}_S) \le |\text{gen}_S(\widehat{f}_S)| \le \Delta$.

Step 4. Steps 2 and 3 in (5) give $\varepsilon_{\text{est}} \le 2\Delta + \varepsilon_{\text{opt}}$; insert this into (2). ■

These are the three error sources of Chapter 8, Section 1.

1.3 A fixed predictor versus a learned predictor

Theorem 8S.7: fixed predictor

Fix f before seeing the sample and let $Z_i = \ell(f(X_i), Y_i)$ have variance $0 < \sigma_f^2 < \infty$. Then

$$\mathbb{E}_S[\widehat{R}_S(f) - R(f)] = 0, \qquad \text{Var}_S\left(\widehat{R}_S(f) - R(f)\right) = \frac{\sigma_f^2}{n},$$

and for every $c > 0$

$$P_S\Big(|\widehat{R}_S(f) - R(f)| \ge \frac{c\,\sigma_f}{\sqrt{n}}\Big) \le \frac{1}{c^2}.$$

Hence $\widehat{R}_S(f) - R(f) = O_P(n^{-1/2})$. If $\sigma_f = 0$, the difference is 0 almost surely.

Proof. Step 1 (iid losses). f is free of S, so the Z_i are iid with mean $R(f)$, variance σ_f^2.

Step 2 (mean and variance). $\widehat{R}_S(f) = \frac{1}{n}\sum_i Z_i$ has mean $R(f)$ and variance $\frac{1}{n^2}\cdot n\sigma_f^2 = \sigma_f^2/n$.

Step 3 (Chebyshev). $P(|D| \ge t) \le v/t^2$ with $v = \sigma_f^2/n$, $t = c\sigma_f/\sqrt{n}$.

Step 4 ($\sigma_f = 0$). $Z_i = R(f)$ a.s., so $\widehat{R}_S(f) = R(f)$ a.s. ■

This fails for $\widehat{f}_S$: its losses $\ell(\widehat{f}_S(X_i), Y_i)$ use a predictor chosen from the same sample, so they are neither independent nor centred at $R(\widehat{f}_S)$.

Example 8S.8: a memorizing predictor [Supplement]

Let X be uniform on $[0,1]$ and $Y \in \{-1,1\}$ a fair coin independent of X; use the 0–1 loss. Every predictor has $R(f) = \frac{1}{2}$. Let $\widehat{f}_S(x) = Y_i$ if $x = X_i$ and $\widehat{f}_S(x) = 1$ otherwise. Then $\widehat{R}_S(\widehat{f}_S) = 0$, but a fresh X a.s. misses every X_i, so $R(\widehat{f}_S) = \frac{1}{2}$: the gap $\frac{1}{2}$ does not shrink with n, though every *fixed* f has gap $O_P(n^{-1/2})$ (Theorem 8S.7). Uniform control $\Delta_S(\mathcal{F}_\delta)$ fails because the class is too rich.

Expectation and high-probability statements. Guarantees take the forms

$$\mathbb{E}_S|\text{gen}_S(\widehat{f}_S)| \le \varepsilon \qquad \text{and} \qquad P_S\{|\text{gen}_S(\widehat{f}_S)| \le \varepsilon\} \ge 1 - \eta.$$

(forms, not consequences). The second is the PAC viewpoint (a full definition also fixes the target and the growth of n). A small *signed* $\mathbb{E}_S \operatorname{gen}_S(\widehat{f}_S)$ implies neither: gaps can cancel.

Theorem 8S.9: complexity trade-off for nested classes

If $\mathcal{F}_\delta \subseteq \mathcal{F}_{\delta'} \subseteq \mathcal{F}$, then $\varepsilon_{\text{app}}(\mathcal{F}_{\delta'}) \le \varepsilon_{\text{app}}(\mathcal{F}_\delta)$ and $\Delta_S(\mathcal{F}_{\delta'}) \ge \Delta_S(\mathcal{F}_\delta)$.

Proof. Infima over larger sets are smaller, suprema larger. ■

Enlarging the class trades approximation error for uniform deviation (a statement about bounds, not test-error curves).

2 The Curse of Dimensionality

Motivation. A linear target needs about d samples, a Lipschitz one exponentially many.

Example 8S.10: noiseless linear recovery

Let $y_i = \mathbf{x}_i^\top \boldsymbol{\theta}^*$ with design matrix $\mathbf{X} \in \mathbb{R}^{n\times d}$ (rows $\mathbf{x}_i^\top$). Then $\boldsymbol{\theta}^*$ is uniquely determined by the data if and only if $\operatorname{rank} \mathbf{X} = d$. In particular $n \ge d$ is necessary, but not sufficient by itself.

Proof. Step 1 (sufficiency). Independent columns: $\mathbf{X}\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\theta}' \Rightarrow \boldsymbol{\theta} = \boldsymbol{\theta}'$.

Step 2 (necessity). If $\operatorname{rank} \mathbf{X} < d$, some $\mathbf{v} \neq \mathbf{0}$ has $\mathbf{X}\mathbf{v} = \mathbf{0}$, and $\boldsymbol{\theta}^* + \mathbf{v}$ fits equally well.

Step 3. $\operatorname{rank} \mathbf{X} \le \min(n, d)$; a repeated row shows $n \ge d$ is not enough. ■

Theorem 8S.11: Gaussian designs have full rank [Supplement]

If the rows $\mathbf{x}_i$ are iid $\mathcal{N}(\mathbf{0}, \Sigma)$ with Σ positive definite and $n \ge d$, then $\operatorname{rank} \mathbf{X} = d$ almost surely.

Proof. Step 1. It suffices that the top block has $\det \mathbf{X}_d \neq 0$.

Step 2. $\det \mathbf{X}_d$ is a nonzero polynomial in d^2 entries ($= 1$ at $\mathbf{I}$); its zero set is Lebesgue-null.

Step 3. $\Sigma \succ 0$ gives the entries a joint density, so they avoid null sets a.s. ■

2.1 Lipschitz targets and nearest neighbours

Let $y_i = f^*(\mathbf{x}_i)$, $\operatorname{Lip}(f^*) \le \beta$ ($\operatorname{Lip}(f) = \sup_{\mathbf{x}\neq\mathbf{x}'} |f(\mathbf{x}) - f(\mathbf{x}')|/\|\mathbf{x} - \mathbf{x}'\|$), and a minimum-Lipschitz interpolant

$$\widehat{f} \in \arg\min \{\operatorname{Lip}(f) : \ f(\mathbf{x}_i) = y_i \text{ for all } i\}. \tag{6}$$

Theorem 8S.12: error controlled by the nearest observation

If $\mathbf{x}_k$ is a training point nearest to $\mathbf{x}$, then $|\widehat{f}(\mathbf{x}) - f^*(\mathbf{x})| \le 2\beta\|\mathbf{x} - \mathbf{x}_k\|$.

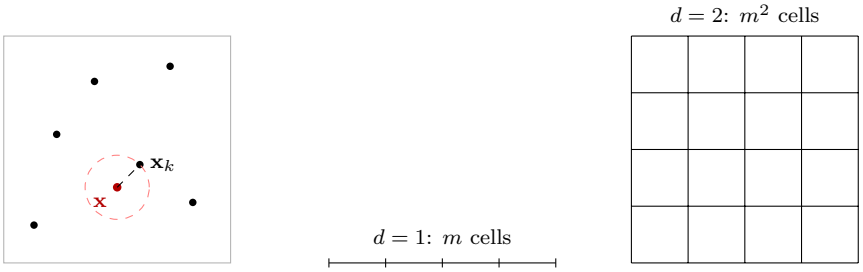
Proof. Step 1. f^* is feasible in (6), so $\operatorname{Lip}(\widehat{f}) \le \operatorname{Lip}(f^*) \le \beta$.

Step 2. Since $\widehat{f}(\mathbf{x}_k) = y_k = f^*(\mathbf{x}_k)$,

$$|\widehat{f}(\mathbf{x}) - f^*(\mathbf{x})| \le |\widehat{f}(\mathbf{x}) - \widehat{f}(\mathbf{x}_k)| + |f^*(\mathbf{x}_k) - f^*(\mathbf{x})| \le \beta\|\mathbf{x} - \mathbf{x}_k\| + \beta\|\mathbf{x} - \mathbf{x}_k\|.$$

■

Resolution $1/m$ on $[0,1]^d$ needs one point in each of m^d cells: $n \approx m^d$, $\log n \approx d \log m$.



(a) local coverage (b) fixed resolution $1/m$

Fixed resolution costs m^d cells.

Theorem 8S.13: nearest-neighbour distance in the unit cube [Supplement]

Let $X, X_1, \dots, X_n$ be iid uniform on $[0,1]^d$ and $D = \min_i \|X - X_i\|$. Then

$$\mathbb{E}\, D^2 \le C_d\, n^{-2/d}, \qquad C_d = d\,\Gamma(1 + \tfrac{2}{d}).$$

Consequently $\mathbb{E}|\widehat{f}(X) - f^*(X)|^2 \le 4\beta^2 C_d n^{-2/d}$, and this bound is at most ε once $n \ge (4\beta^2 C_d/\varepsilon)^{d/2}$.

Proof. Step 1 (small-ball probability). For $0 < r \le \sqrt{d}$ the box $\{\mathbf{u} \in [0,1]^d : |u_j - x_j| \le r/\sqrt{d}\ \forall j\}$ lies in the r-ball ($\|\mathbf{u} - \mathbf{x}\| \le \sqrt{d} \max_j |u_j - x_j|$) and has sides $\ge r/\sqrt{d}$, so

$$p(\mathbf{x}, r) := P(\|X_1 - \mathbf{x}\| \le r) \ge d^{-d/2} r^d.$$

Step 2 (tail of D). By conditional independence and $1 - p \le e^{-p}$,

$$P(D > r \mid X = \mathbf{x}) = (1 - p(\mathbf{x}, r))^n \le \exp\big(-n\, d^{-d/2} r^d\big) \qquad (0 < r \le \sqrt{d}).$$

For $r > \sqrt{d}$ (the diameter) the left side is 0.

Step 3 (integrate the tail).

$$\mathbb{E}[D^2 \mid X] \le \int_0^\infty \exp\big(-n\, d^{-d/2} s^{d/2}\big)\, ds.$$

Substitute $u = n\, d^{-d/2} s^{d/2}$, i.e. $s = d\,(u/n)^{2/d}$ and $ds = d\, n^{-2/d} \frac{2}{d}\, u^{2/d-1} du$:

$$\int_0^\infty e^{-u}\, d\, n^{-2/d} \tfrac{2}{d}\, u^{2/d-1}\, du = d\, n^{-2/d}\, \tfrac{2}{d}\, \Gamma(\tfrac{2}{d}) = d\,\Gamma(1 + \tfrac{2}{d})\, n^{-2/d}.$$

Take the expectation over X.

Step 4 (prediction error). Square Theorem 8S.12 at $\mathbf{x} = X$, take expectations, solve for n. ■

Squared error ε needs $n \sim \varepsilon^{-d/2}$ (ε^2: ε^{-d}).

Remark 8S.14: upper versus lower bounds

This uses uniform inputs, and an upper bound gives only a *sufficient* sample size, not a lower bound for all algorithms.

3 Surrogate Loss Is Not Classification Error

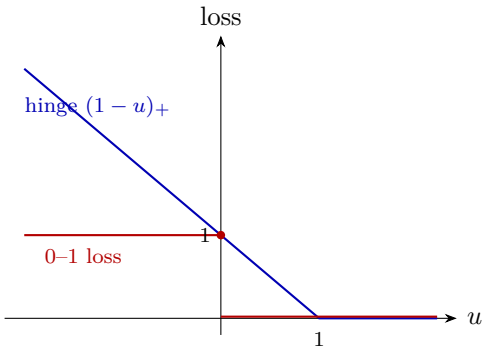
Motivation. Small hinge risk need not mean fewest mistakes.

$Y \in \{-1, 1\}$, score g, prediction sign g, **signed margin** $u = Yg(X)$:

$$\ell_{\text{hinge}}(u) = (1-u)_+ = \max\{0, 1-u\}, \qquad \mathbf{1}\{u \le 0\} \le (1-u)_+. \tag{7}$$

Proof of the inequality. If $u \le 0$, then $1 - u \ge 1 = \mathbf{1}\{u \le 0\}$. If $u > 0$, the left side is $0 \le (1-u)_+$. ■

Any tie-breaking gives a misclassification indicator $\le \mathbf{1}\{u \le 0\} \le$ hinge.



Hinge also penalizes correct predictions with margin below 1.

Example 8S.15: different rankings

Let $P(X = x_1) = 0.99$, $P(X = x_2) = 0.01$, $Y = +1$:

	$g(x_1)$	$g(x_2)$	R_{0-1}	R_{hinge}
g_A	0.1	0.1	0	0.9
g_B	1	−0.1	0.01	0.011

g_A makes no mistakes, yet hinge risk prefers g_B.

Computation. g_A: margins $0.1 > 0$, no errors, hinge 0.9 each. g_B: x_1 margin 1, hinge 0; x_2 margin -0.1, hinge 1.1, wrong. So $R_{0-1}(g_B) = 0.01$, $R_{\text{hinge}}(g_B) = 0.01 \cdot 1.1 = 0.011$. ■

Remark 8S.16: no contradiction

$R_{0-1} \le R_{\text{hinge}}$ holds for each score ($0 \le 0.9$, $0.01 \le 0.011$); it does not order different scores. Controlling hinge risk does not pick the class member with the least classification error; classification-calibration results concern minimizers over all measurable scores.

4 Universal Approximation

Motivation. Can one hidden layer approximate any continuous target? This controls ε_{app} in (2).

$K = [0,1]^d$, $C(K)$ with $\|f\|_\infty = \sup_{\mathbf{x}\in K} |f(\mathbf{x})|$ (Banach). U is **dense** if it approximates every $f \in C(K)$ arbitrarily well.

Definition 8S.17: sigmoidal activation

A continuous $\sigma : \mathbb{R} \to [0,1]$ is **sigmoidal** if $\lim_{t\to-\infty} \sigma(t) = 0$ and $\lim_{t\to+\infty} \sigma(t) = 1$.

(The handwritten definition asks for C^1; continuity suffices.)

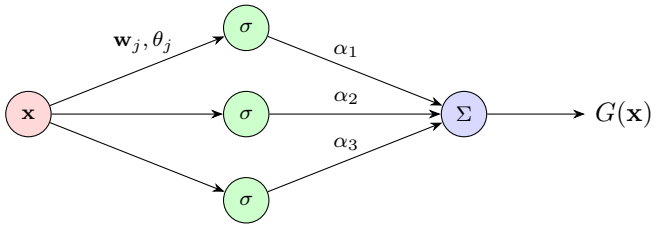
Theorem 8S.18: Cybenko 1989

For a fixed continuous sigmoidal σ, the class

$$\mathcal{G}_\sigma = \Big\{ \mathbf{x} \mapsto \sum_{j=1}^{N} \alpha_j \sigma(\mathbf{w}_j^\top \mathbf{x} + \theta_j) : \ N \in \mathbb{N},\ \alpha_j, \theta_j \in \mathbb{R},\ \mathbf{w}_j \in \mathbb{R}^d \Big\}$$

is dense in $(C(K), \|\cdot\|_\infty)$: for every $f \in C(K)$ and $\varepsilon > 0$ there is $G \in \mathcal{G}_\sigma$ with $\sup_{\mathbf{x}\in K} |f(\mathbf{x}) - G(\mathbf{x})| < \varepsilon$.

The proof is in Section 5 (and in Chapter 8, Theorem 8.4).



One hidden layer ("two-layer"); the width varies.

Quantifiers. N and the weights depend on f, ε: no fixed width, no bound on N, no algorithm.

Theorem 8S.19: link to approximation error

Let $f^* \in C(K)$, $Y = f^*(X)$ with X supported in K, and squared loss. If $\sup_K |f^* - G| < \varepsilon$, then $R(G) = \mathbb{E}(G(X) - f^*(X))^2 < \varepsilon^2$. Hence $\varepsilon_{\text{app}}(\mathcal{G}_\sigma) = 0$ for this problem.

Proof. Pointwise $(G(X) - f^*(X))^2 < \varepsilon^2$ on K; take expectations. As $R \geq 0 = R(f^*)$, $\varepsilon \to 0$ gives $\inf_{\mathcal{G}_\sigma} R = 0$. ■

(This uses the squared loss.)

5 The Functional-Analytic Proof

Motivation. Failure of density yields a nonzero measure blind to every neuron, which cannot exist.

$M(K)$: finite regular signed Borel measures, $|\mu|$ total variation. Integration against μ is continuous linear:

$$\Big| \int_K h \, d\mu \Big| \leq \|h\|_\infty \, |\mu|(K). \tag{8}$$

Definition 8S.20: discriminatory activation

σ is **discriminatory** on K if for every $\mu \in M(K)$

$$\int_K \sigma(\mathbf{w}^\top \mathbf{x} + \theta) \, d\mu(\mathbf{x}) = 0 \quad \text{for all } \mathbf{w}, \theta \qquad \Longrightarrow \qquad \mu = 0.$$

Theorem 8S.21: two separation tools

(i) If a linear subspace U is not dense in a normed space X, there is a bounded linear functional $L \neq 0$ on X with $L|_U = 0$.

(ii) (Riesz) Every bounded linear functional L on $C(K)$ has a unique representation $L(h) = \int_K h\, d\mu$ with $\mu \in M(K)$, and $\|L\| = |\mu|(K)$.

Proof. **(i)** [Supplement] (as Chapter 8, Theorem 8.5). Pick $x_0 \notin \overline{U}$, $\rho = \inf_{u\in\overline{U}}\|x_0 - u\| > 0$; $\ell(u + tx_0) = t$ on $\overline{U} \oplus \mathbb{R}x_0$ vanishes on U, $\ell(x_0) = 1$, $\|\ell\| \leq 1/\rho$ by $|t|\rho \leq \|u + tx_0\|$; extend by Hahn–Banach.

(ii) Riesz representation, quoted without proof. ■

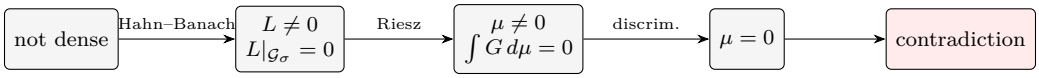
Theorem 8S.22: discriminatory activations give density

If σ is continuous and discriminatory, then $\mathcal{G}_\sigma$ is dense in $C(K)$.

Proof. Step 1 (subspace). $\mathcal{G}_\sigma \subseteq C(K)$, closed under sums (concatenate, N varies) and scalars.

Step 2 (a nonzero measure). If not dense, Theorem 8S.21(i) gives $L \neq 0$, $L|_{\mathcal{G}_\sigma} = 0$, and (ii) gives $\mu \neq 0$ with $L(h) = \int h\, d\mu$.

Step 3 (contradiction). Single units lie in $\mathcal{G}_\sigma$, so $\int_K \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu = 0$ for all $\mathbf{w}, \theta$, forcing $\mu = 0$. ■



A separation argument, not a construction.

Theorem 8S.23: continuous sigmoidals are discriminatory [Supplement]

Every continuous sigmoidal σ is discriminatory on K.

With Theorem 8S.22 this proves Cybenko's theorem (lecture: no proof; cf. Chapter 8, Theorem 8.9).

Proof. Assume $\int_K \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu = 0$ for all $\mathbf{w}, \theta$.

Step 1 (sharpening). $\Pi = \{\mathbf{x} \in K : \mathbf{w}^\top \mathbf{x} + \theta > 0\}$, $H = \{\mathbf{x} \in K : \mathbf{w}^\top \mathbf{x} + \theta = 0\}$. As $\lambda \to \infty$,

$$\sigma(\lambda(\mathbf{w}^\top \mathbf{x} + \theta) + \varphi) \longrightarrow \begin{cases} 1, & \mathbf{x} \in \Pi, \\ \sigma(\varphi), & \mathbf{x} \in H, \\ 0, & \text{otherwise.} \end{cases}$$

and each left side is a neuron with integral 0.

Step 2 (limit). Dominated convergence (bound 1, on $\mu^\pm$):

$$0 = \mu(\Pi) + \sigma(\varphi)\, \mu(H) \qquad \text{for every } \varphi.$$

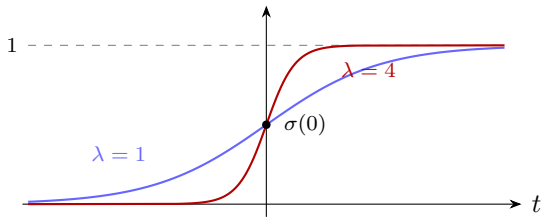
Step 3 (separate the two terms). $\varphi \to -\infty$: $\mu(\Pi) = 0$; $\varphi \to +\infty$: $\mu(H) = 0$.

Step 4 (pushforward). $\nu_{\mathbf{w}}(B) = \mu(\{\mathbf{x} : \mathbf{w}^\top \mathbf{x} \in B\})$ vanishes on half-lines (a, ∞), points, and $(a, b] = (a, \infty) \setminus (b, \infty)$. These with $\mathbb{R}$ form a generating π-system on which $\nu_{\mathbf{w}}^{\pm}$ agree; by π–λ, $\nu_{\mathbf{w}} = 0$.

Step 5 (Fourier). For all $t, \mathbf{w}$,

$$\int_K e^{it\mathbf{w}^\top \mathbf{x}}\, d\mu(\mathbf{x}) = \int_{\mathbb{R}} e^{its}\, d\nu_{\mathbf{w}}(s) = 0.$$

so $\hat{\mu} \equiv 0$ and $\mu = 0$ by Fourier uniqueness. ■



$\sigma(\lambda t)$ sharpens to a step; the value at 0 stays $\sigma(0)$.

Remark 8S.24: two technical points

(1) Use $\varphi \to -\infty$, not a finite φ with $\sigma(\varphi) = 0$ (none exists for the logistic). (2) H may carry signed mass (a point mass); the proof keeps $\sigma(\varphi)\mu(H)$. The handout swaps H and Π; we follow Chapter 8. Baire and Borel coincide on $[0,1]^d$.

6 Fourier Regularity and Barron Classes

Motivation. A rate in the width needs Fourier structure on the target.

$\widehat{f}(\boldsymbol{\xi}) = \int_{\mathbb{R}^d} f(\mathbf{x}) e^{-2\pi i \mathbf{x}^\top \boldsymbol{\xi}}\, d\mathbf{x}$ (here the hat is a Fourier transform, not a predictor).

Definition 8S.25: Sobolev space

For $s \geq 0$,

$$H^s(\mathbb{R}^d) = \Big\{ f \in L^2(\mathbb{R}^d) : \int_{\mathbb{R}^d} (1 + \|\boldsymbol{\xi}\|^2)^s |\widehat{f}(\boldsymbol{\xi})|^2 \, d\boldsymbol{\xi} < \infty \Big\},$$

(Fourier transform in the L^2 sense; $H^0 = L^2$.)

Barron replaces this weighted L^2 control by weighted L^1 control.

Definition 8S.26: Barron class (lecture version)

For $C_1 > 0$,

$$\Gamma_{C_1} = \Big\{ f \in L^1(\mathbb{R}^d) : \ \widehat{f} \in L^1(\mathbb{R}^d), \ \int_{\mathbb{R}^d} 2\pi \|\boldsymbol{\xi}\| \, |\widehat{f}(\boldsymbol{\xi})| \, d\boldsymbol{\xi} \leq C_1 \Big\}.$$

Targets are real, continuous representatives.

On a domain Ω, the **spectral Barron norm** uses extensions:

$$\|f\|_{\mathcal{B}^s(\Omega)} = \inf_{f_e|_\Omega = f} \int_{\mathbb{R}^d} (1 + \|\boldsymbol{\xi}\|)^s |\widehat{f}_e(\boldsymbol{\xi})| \, d\boldsymbol{\xi},$$

over admissible extensions f_e (the cheapest wins); the space is $\{f : \text{norm} < \infty\}$.

Remark 8S.27: convention check

$2\pi\|\boldsymbol{\xi}\| \neq (1 + \|\boldsymbol{\xi}\|)^s$, so Γ_{C_1} is not simply a $\mathcal{B}^1$ ball. Chapter 8 uses strict $< C$; this changes nothing in Theorem 8S.28. Extension conventions vary.

7 A Quantitative Approximation Theorem

Motivation. An explicit rate in N with exponent free of d.

Let $B_1^d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \le 1\}$ and

$$\|h\|^2_{L^2(\mathrm{Unif}(B_1^d))} = \frac{1}{\mathrm{Leb}_d(B_1^d)} \int_{B_1^d} |h(\mathbf{x})|^2 \, d\mathbf{x}.$$

Theorem 8S.28: Barron approximation, lecture statement

Let $f \in \Gamma_{C_1}$ and let σ be continuous and sigmoidal. For every $c > 4C_1^2$ and $N \in \mathbb{N}$ there is a network

$$\Phi_N(\mathbf{x}) = a_0 + \sum_{j=1}^{N} a_j \, \sigma(\mathbf{w}_j^\top \mathbf{x} + b_j)$$

with N hidden units and an output constant a_0 such that $\|f - \Phi_N\|^2_{L^2(\mathrm{Unif}(B_1^d))} < c/N$.

Proof. Step 1 (reduce to Chapter 8). Pick $C_1 < C < \sqrt{c}/2$: $f \in \Gamma_C$ of Chapter 8 and $c > 4C^2$.

Step 2 (apply the full proof). Chapter 8 (Theorem 8.12, Lemmas 8.15, 8.16) gives $f_N(\mathbf{x}) = f(\mathbf{0}) + \sum_{i=1}^{N} \frac{\gamma_i}{N} \sigma(\lambda \mathbf{a}_i^\top \mathbf{x} + \lambda b_i)$ with error $< c/N$, of the stated form with $a_0 = f(\mathbf{0})$, $a_j = \gamma_j/N$, $\mathbf{w}_j = \lambda \mathbf{a}_j$, $b_j \leftarrow \lambda b_j$. ■

Remark 8S.29: the output constant

The proof approximates $f - f(\mathbf{0})$ and restores $f(\mathbf{0})$ as output bias a_0; as an extra hidden unit it would count $N + 1$.

What the rate means. Squared error $O(N^{-1})$, L^2 error $O(N^{-1/2})$, exponent free of d; but the Barron constant may grow with d, and nothing is said about n or gradient descent.

	Universal approximation	Barron approximation
Target class	continuous functions on K	controlled Fourier cost Γ_{C_1}
Error criterion	uniform norm	normalized squared L^2 on B_1^d
Conclusion	arbitrarily accurate	explicit c/N rate
Width	no quantitative bound	quantitative dependence on N

Neither replaces the statistical and optimization terms of Theorem 8S.6. The convex-hull lemma, proof and Rademacher bounds are in Chapter 8.

8 Exercises

Exercise 8S.1 (recovering the decomposition)

Starting from $R(\widehat{f}_S) - \inf_{\mathcal{F}_\delta} R$, derive the three terms of (5). Which two are bounded in absolute value by $\Delta_S(\mathcal{F}_\delta)$?

Solution. Add and subtract $\widehat{R}_S(\widehat{f}_S)$ and $\inf_{\mathcal{F}_\delta} \widehat{R}_S$:

$$R(\widehat{f}_S) - \inf R = [R(\widehat{f}_S) - \widehat{R}_S(\widehat{f}_S)] + [\widehat{R}_S(\widehat{f}_S) - \inf \widehat{R}_S] + [\inf \widehat{R}_S - \inf R].$$

The first (gap) and third (Steps 1–2 of Theorem 8S.6) are bounded by Δ_S; the middle is $\varepsilon_{\mathrm{opt}}$.

Exercise 8S.2 (fixed versus learned)

Why can the variance calculation of Theorem 8S.7 not be reused for $\widehat{f}_S$?
Solution. It needs iid Z_i with mean $R(f)$, i.e. f independent of S; each $\ell(\widehat{f}_S(X_i), Y_i)$ depends on the whole sample, and Example 8S.8 keeps the gap at $\frac{1}{2}$.

Exercise 8S.3 (hinge ordering)

In Example 8S.15, why is the ordering $R_{\text{hinge}}(g_B) < R_{\text{hinge}}(g_A)$ consistent with $R_{0\text{–}1} \le R_{\text{hinge}}$?
Solution. The bound compares each score's own risks ($0 \le 0.9$, $0.01 \le 0.011$), not *different* scores.

Exercise 8S.4 (fixed width)

Why is $\mathcal{G}_\sigma$ (all widths) a linear space while the class of width exactly N need not be?
Solution. A sum of width-N networks has width $2N$: for $N = 1$, logistic σ, $\sigma(t) + \sigma(t-5)$ is not $\alpha\sigma(wt+\theta)$ (constant or one inflection point), since its second derivative changes sign three times (+ at $t = 0, 4$, − at $t = 1, 6$). All widths together are closed under addition.

Exercise 8S.5 (where each ingredient enters)

Point to where $L \neq 0$, $\mu \neq 0$ and the discriminatory property enter the density proof.
Solution. Step 2: $L \neq 0$, $L|_{\mathcal{G}_\sigma} = 0$ (Theorem 8S.21(i)), then $\mu \neq 0$ by Riesz (ii); Step 3: discrimination gives $\mu = 0$.

Exercise 8S.6 (four symbols)

Distinguish n, N, N^{-1} and $N^{-1/2}$.
Solution. n: sample size ($O_P(n^{-1/2})$ for a fixed predictor); N: width; N^{-1}: Barron's *squared* L^2 rate; $N^{-1/2}$: the L^2 rate.